Tallinn University of Technology
Department of Electrical Power Engineering and Mechatronics

ATR0030 Robotics

Lab 3. ABB IRB 1600 robot

Running and correcting the circle and "JB" drawing program on the real robot

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Task

The task of this work was to run the RAPID program prepared in Lab 2 on the real ABB IRB 1600 robot, so that it draws a circle and my initials (JB) on a sheet of paper with a marker. The robot controller was connected to ABB RobotStudio, the program was loaded and run from the FlexPendant, and the program was corrected step by step until the drawing came out clean. Five test runs were made: step by step and continuous at 25 % speed, and finally at 100 % speed in automatic mode.

a. Robot name, degrees of freedom, body names and joint types

Robot name	ABB IRB 1600-6/1.45
Degrees of freedom	6
Body names	Base, Link1, Link2, Link3, Link4, Link5, Link6
Joint types	6 revolute joints

b. Kinematic scheme of the robot

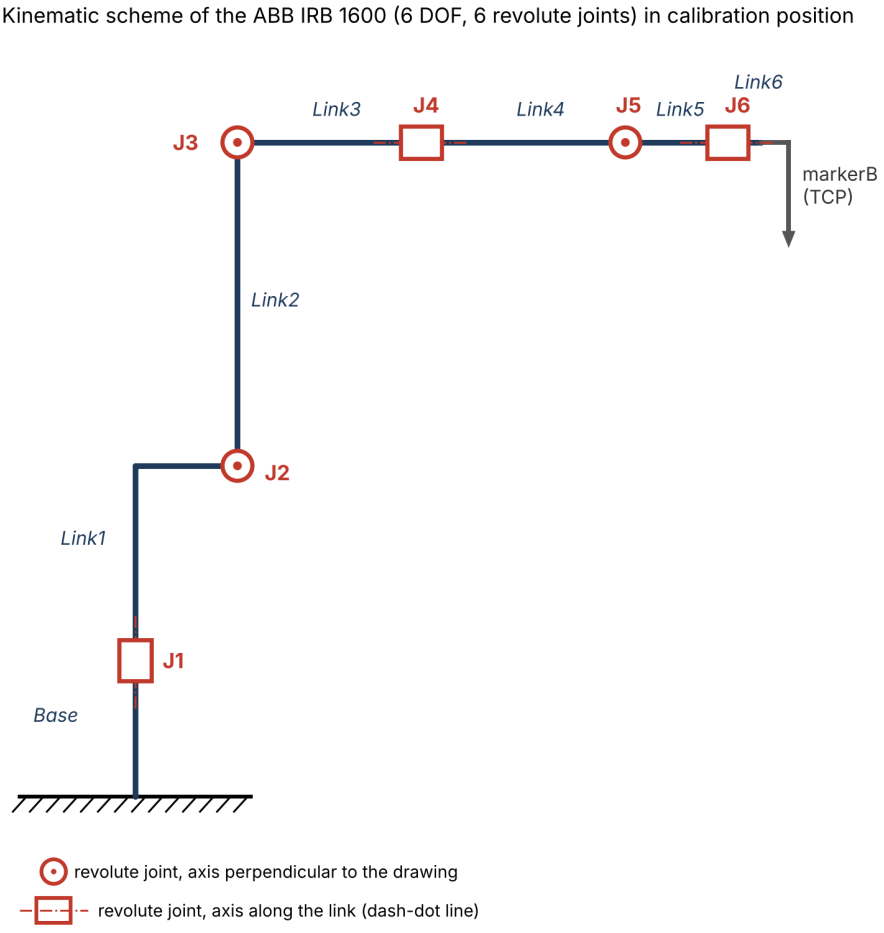


Figure 1: Kinematic scheme of the IRB 1600. J1 rotates the whole arm about the vertical axis, J2 and J3 move the lower and upper arm, J4 rolls the upper arm, J5 bends the wrist and J6 rotates the tool flange carrying markerB.

c. Executable RAPID program with explanation of corrections made

The program from Lab 2 was loaded on the real robot and corrected after each test run. Table 1 lists the corrections, and the final program is given in Listing 1. The corrected lines are marked with `! CORRECTED` or `! NEW` comments.

Change	Problem seen on the paper	Why the change fixes it
New targets `Target_50up` and `Target_90up`, moves to them at the end of Path_10 and Path_20	A "skid mark" line from the end of the circle (Target_50) to the start of the J (Target_60), and from the end of the J (Target_90) to the B.	The pen is lifted 10 mm above the paper at the end of each shape, so it no longer drags across the page while moving to the next shape.
Zones in Path_10 changed from `z100` to `fine`	The start and end of the circle were cut off and the circle was not closed; the robot cut the corner straight towards the next point.	With `z100` the robot may pass up to 100 mm away from a target. With `fine` it stops exactly on each target, so the circle starts and closes in the right place.
Letter moves changed from `MoveJ` to `MoveL`	Strokes of the letters were not straight and the shapes were distorted.	`MoveJ` only controls the joints, so the pen path between targets is not straight. `MoveL` moves the pen in a straight line.
Zones on the curved parts of the J and B changed from `z0` to `z50`	The hook of the J and the bowls of the B were made of straight lines with sharp corners.	A larger zone lets the robot blend the corners between neighbouring targets, so the curves come out rounded.

Table 1: Corrections made to the program.

Listing 1: Final Module1 run on the real robot (RAPID).

```
MODULE Module1
    ! ATR0030 Lab 3 - Jordan Brodie - final program run on the real IRB 1600
    ! Draws a circle (Path_10) and the initials J (Path_20) and B (Path_30) on Paper1 with markerB.
    ! Target orientation [0,0.707,0.707,0] = marker pointing straight down; config [-1,-1,0,1] = Cfg3.
    ! Z = 0 -> pen on paper, Z = 10 -> pen lifted (UP targets)

    ! --- Circle targets (AutoPath around cylinder r = 70 mm, centre 148.5,105) ---
    CONST robtarget Target_10:=[[148.5,35,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_20:=[[218.5,105,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_30:=[[148.5,175,0.1],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_40:=[[78.5,105,0.1],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_50:=[[148.5,35,0.1],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_50up:=[[148.5,35,10],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]]; ! NEW: lift after circle

    ! --- Letter J targets ---
    CONST robtarget Target_60:=[[96.373,122.697,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_60up:=[[96.373,122.697,10],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_70:=[[128.291,123.702,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_80:=[[127.45,85.334,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_90:=[[99.327,88.197,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
    CONST robtarget Target_90up:=[[99.327,88.197,10],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
        9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]]; ! NEW: lift after J

    ! --- Letter B targets ---
```

```
CONST robtarget Target_100:=[[153.65,126.951,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_100up:=[[153.65,126.951,10],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_110:=[[176.968,128.117,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_120:=[[169.847,106.303,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_140:=[[169.849,106.04,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_130:=[[156.283,105.762,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_150:=[[172.387,86.337,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_160:=[[158.109,85.033,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_170:=[[154.681,128.259,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
! --- Extra points that shape the J (220-240) and the B (250, 180-210) ---
CONST robtarget Target_220:=[[129.862,99.321,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_230:=[[120.736,81.348,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_240:=[[107.716,81.075,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_250:=[[179.813,123.004,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_180:=[[180.068,115.515,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_190:=[[176.304,110.207,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_200:=[[177.546,99.997,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];
CONST robtarget Target_210:=[[179.316,92.92,0],[0,0.707106781,0.707106781,0],[-1,-1,0,1],[
    9E+09,9E+09,9E+09,9E+09,9E+09,9E+09]];

PROC main() ! Jordan Brodie entry point
    Path_10;   ! draw the circle
    Path_20;   ! draw the letter J
    Path_30;   ! draw the letter B
    MoveAbsJ calib_pos,v1000,z0,tool0;   ! return to calibration position (all joints 0)
    Stop;   ! end of program
ENDPROC

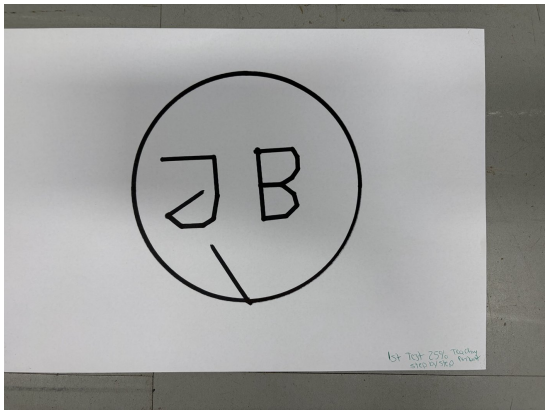
PROC Path_10() ! circle: two circular moves through four points
    MoveJ Target_10,v1000,fine,markerB\WObj:=Paper1;            ! CORRECTED z100 -> fine (exact start)
    MoveC Target_20,Target_30,v1000,fine,markerB\WObj:=Paper1;  ! CORRECTED z100 -> fine
    MoveC Target_40,Target_50,v1000,fine,markerB\WObj:=Paper1;  ! CORRECTED z100 -> fine (circle closes)
    MoveL Target_50up,v1000,z100,markerB\WObj:=Paper1;          ! NEW: pen up, no skid mark to the J
ENDPROC

PROC Path_20() ! letter J
    MoveJ Target_60up,v1000,z0,markerB\WObj:=Paper1;   ! hover 10 mm above start of J
    MoveL Target_60,v1000,z0,markerB\WObj:=Paper1;     ! CORRECTED MoveJ -> MoveL, pen down
    MoveL Target_70,v1000,z0,markerB\WObj:=Paper1;     ! top bar
    MoveL Target_220,v1000,z50,markerB\WObj:=Paper1;   ! stem, CORRECTED z0 -> z50 (rounded hook)
    MoveL Target_80,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_230,v1000,z50,markerB\WObj:=Paper1;   ! hook
    MoveL Target_240,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_90,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_90up,v1000,z50,markerB\WObj:=Paper1;  ! NEW: pen up, no skid mark to the B
ENDPROC

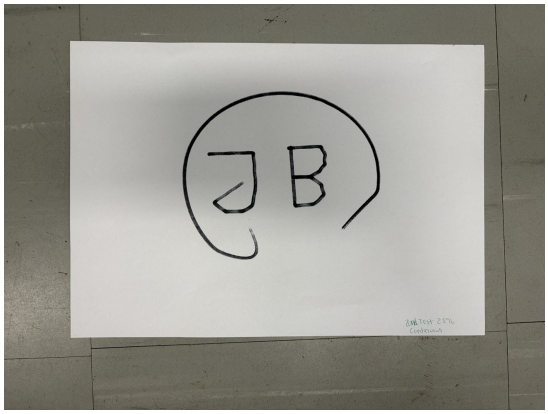
PROC Path_30() ! letter B
    MoveJ Target_100up,v1000,z0,markerB\WObj:=Paper1;  ! hover 10 mm above start of B
    MoveL Target_100,v1000,z0,markerB\WObj:=Paper1;    ! CORRECTED MoveJ -> MoveL, pen down
    MoveL Target_110,v1000,z50,markerB\WObj:=Paper1;   ! top, CORRECTED z0 -> z50 (rounded bowls)
    MoveL Target_250,v1000,z50,markerB\WObj:=Paper1;   ! upper bowl
    MoveL Target_180,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_190,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_120,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_140,v1000,z0,markerB\WObj:=Paper1;    ! sharp corner kept (z0)
    MoveL Target_130,v1000,z0,markerB\WObj:=Paper1;    ! middle bar to stem
    MoveL Target_140,v1000,z50,markerB\WObj:=Paper1;
    MoveL Target_200,v1000,z50,markerB\WObj:=Paper1;   ! lower bowl
    MoveL Target_210,v1000,z50,markerB\WObj:=Paper1;
```

```
        MoveL Target_150,v1000,z50,markerB\WObj:=Paper1;
        MoveL Target_160,v1000,z0,markerB\WObj:=Paper1;    ! bottom
        MoveL Target_170,v1000,z0,markerB\WObj:=Paper1;    ! stem back up, closes the B
    ENDPROC
ENDMODULE
```

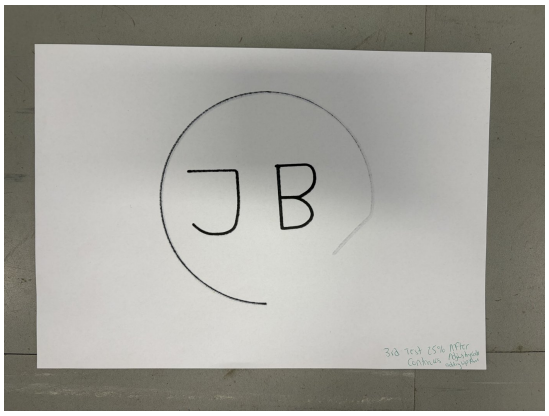
d. Photo of the paths created at the robotic station



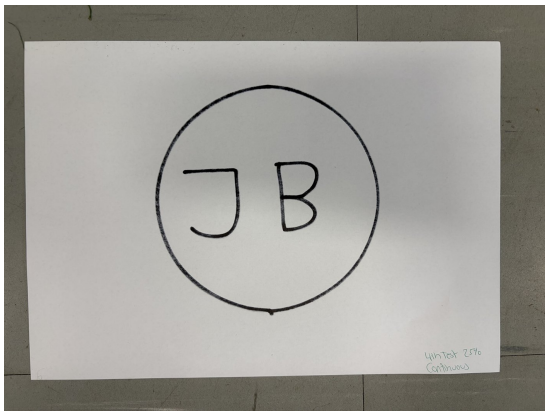
(a) Test 1: 25 %, step by step.



(b) Test 2: 25 %, continuous.



(c) Test 3: 25 %, continuous, after corrections.



(d) Test 4: 25 %, continuous, final program.

Figure 2: Drawings from the test runs, in order.

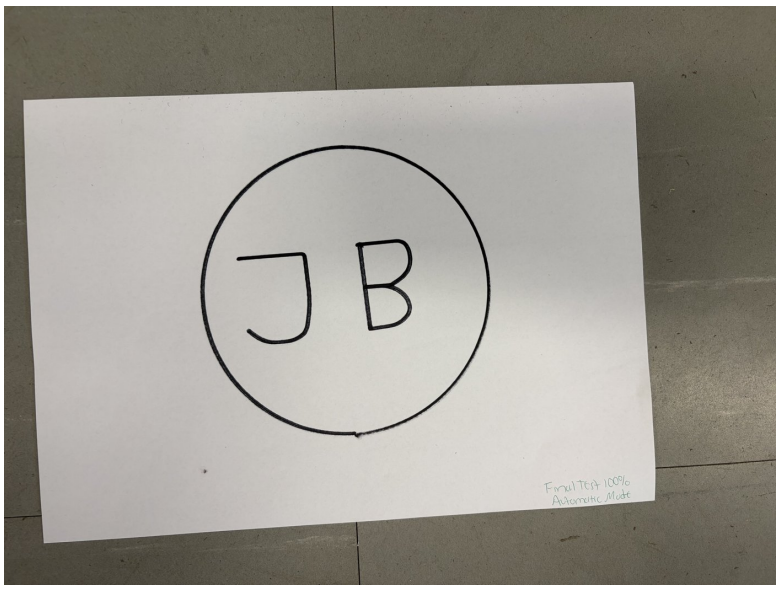


Figure 3: Final result: circle and initials JB drawn by the IRB 1600 at 100 % speed in automatic mode (test 5).

Test	Mode	Result
1	25 %, step by step on the FlexPendant	Circle complete, but skid marks from the end of the circle to the start of the J and from the end of the J to the B. Curves of J and B made of straight segments.
2	25 %, continuous	Start and end of the circle wrong and the circle not closed; curves of J and B still straight segments.
3	25 %, continuous, after corrections	UP targets, `MoveL` and `z50` added: no skid marks and rounded J and B. Circle start still cut off, because the first move of Path_10 still had `z100`.
4	25 %, continuous	Last `z100` on `Target_10` changed to `fine`: full closed circle and clean letters.
5	100 %, automatic mode	Same clean result as test 4 at full speed.

Table 2: Summary of the test runs.

e. Conclusions explaining the corrections made to the program

Running the program on the real robot was very different from running it in simulation in Lab 2. In RobotStudio the circle and the initials JB were drawn correctly, but on the real robot the first tests showed several problems. This showed me that a simulation can only go so far: the real pen, paper and robot motion behave differently, so a program has to be tested and tuned on the real machine. For future work I will expect this and plan time for testing and correcting on the hardware.

The corrections were made step by step. Testing started at 25 % speed, step by step on the FlexPendant, so each move could be checked safely. In the first test the pen stayed on the paper after the circle and after the J, which left a line between the shapes. While writing the program in Lab 2 I had already suspected that each shape should end at an UP point, not only start from one, or the pen would drag to the next shape. The test confirmed this, and adding `Target_50up` and `Target_90up` removed the lines. The lesson for me is to trust this kind of intuition while programming and to act on it before testing.

In the continuous tests the start and end of the circle were wrong, because the zone `z100` let the robot pass up to 100 mm away from the targets. Changing the zones of the circle to `fine` made the robot reach each target exactly; one `z100` that was missed on `Target_10` in test 3 was fixed before test 4. My initials used many targets, so they took more editing than a simple shape. The letter moves were changed from `MoveJ` to `MoveL` for straight strokes, and the zones on the curves were changed from `z0` to `z50`. With `z0` the hook of the J and the bowls of the B were made of straight lines with sharp corners; with `z50` the robot blends the corners and the curves become round. After these corrections the final run at 100 % speed in automatic mode drew a clean, closed circle with smooth initials.

I also noticed that Lab 1 (Mitsubishi RV-2AJ in CIROS Studio) and Labs 2–3 (ABB IRB 1600 in RobotStudio) followed the same workflow, even with different robots and software: define the points (targets), enter them into the software, build a path through them, simulate it, and then send the program to the real robot and correct it. This seems to be a general structure of robot programming, and it is something I would like to explore further in other robotic systems.

Overall, I found this lab and the course very interesting and enjoyable, and it gave me practical experience with the full process of programming an industrial robot, from simulation to a working result on the real machine.

f. Video of the program execution

The execution of the final program on the real robot was recorded and is submitted as a separate file: **JordanBrodie_Lab3.mp4**.